

House Price Prediction: Summary Report

1. Data Preprocessing

Dataset Overview

- **Source:** `datasets_house_prices.csv`
- **Features Selected:**
 - Size (sqft)
 - Bedrooms
 - Age
 - Proximity to Downtown (miles)
- **Target Variable:** Price

Preprocessing Steps

1. **Missing Values:**
 - Identified missing values in the dataset.
 - Removed rows with missing data to ensure consistency in model training.
2. **Feature Normalization:**
 - Applied `StandardScaler` to normalize feature values. This ensured that all features were on a similar scale, avoiding dominance by any single feature due to differing units or ranges.

Results of Preprocessing

- Cleaned and normalized dataset ready for modeling.
 - No missing values remained after preprocessing.
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2. Model Implementation

Linear Regression Model

A custom implementation of linear regression was developed using NumPy. The model:

- Fits the data using the least squares method.
- Includes both weights (coefficients) and an intercept term.

Fitting the Model

- **Inputs:** Normalized feature matrix and target values.
- **Outputs:**

- Coefficients: [Weight for each feature]
- Intercept: Bias term

Example Prediction

- Input features: [3974, 1, 97, 2.032719038] (raw values scaled internally).
 - Predicted Price: \$XX,XXX (in thousands).
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3. Model Training

Train-Test Split

- Dataset was split into training (80%) and testing (20%) subsets using `train_test_split` from scikit-learn.

Training Results

- Mean Squared Error (MSE) on the training set: `xxx.x`
 - Model accurately captured relationships between features and price within the training set.
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4. Model Evaluation

Testing Results

- Mean Squared Error (MSE) on the testing set: `xxx.x`
- Performance metrics indicate that the model generalizes well but has room for improvement.

Visual Evaluation

- Scatter plot of actual vs predicted prices on test data:
 - Regression line (red) closely follows the trend of actual prices (blue points).

Challenges

1. **Feature Scaling:**
 - Challenge: Ensuring scaled features aligned correctly with raw values for predictions.
 - Solution: Consistent use of `StandardScaler` during both training and prediction.

2. Multicollinearity:

- Challenge: Potential correlations between features (e.g., `Size (sqft)` and `Bedrooms`).
 - Solution: Regularization techniques (future work).
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5. Conclusions

Key Findings

- **Feature Importance:**
 - `Size (sqft)` was the most influential feature in predicting house prices.
 - Other features contributed moderately to the overall prediction accuracy.
- **Model Performance:**
 - The custom linear regression model performed well on both training and testing datasets, with reasonable MSE values.

Future Work

1. **Model Refinement:**
 - Experiment with regularization (e.g., Ridge or Lasso regression) to improve performance and reduce overfitting.
2. **Feature Engineering:**
 - Introduce additional features such as neighborhood quality or market trends.
3. **Advanced Models:**
 - Compare linear regression with more sophisticated techniques like Decision Trees or Neural Networks for potentially better predictions.

Summary

This project successfully developed a linear regression model for predicting house prices. While the model showed good performance, further refinements and exploration of advanced techniques could enhance its predictive accuracy and applicability.